

Feeding Activity and Rates of Digestion of Northern Squawfish (*Ptychocheilus oregonensis*)

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Trapnet catches of northern squawfish (*Ptychocheilus oregonensis*) at Griffen and Cultus lakes, British Columbia, indicated peak activity and feeding in the twilight and dark hours. At Griffen Lake, squawfish force-fed 2.0 g of redbreasted sunfish (*Richardsonius balteatus*), and held at 17-20 C, digested at a rate of roughly 14%/h. At Cultus Lake, squawfish trapped during the peak of feeding activity, and held at 10-12 C, were sacrificed at 6-h intervals, and showed a digestion rate of sockeye salmon (*Oncorhynchus nerka*) smolts of 14%/h. Larger fish ate larger volumes of food. Regression techniques are not completely adequate for describing the course of digestion, which apparently involves an initial period of retention of food, followed by a fairly rapid evacuation of most of the ingested material. Most fish cleared their stomach contents in 24 h, and one half had done so between 12 and 18 h. In the laboratory, squawfish force-fed rainbow trout (*Salmo gairdneri*) had rates of digestion which increased with temperature, from about 5%/h at 4-6 C, to 40-50%/h at 24 C. Larger weights of food were digested at slower rates, but there were no consistent differences in rate of digestion by different sizes of squawfish.

The rates of digestion are higher than those reported for various predatory European fishes.

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Les prises de sauvagesses du nord (*Ptychocheilus oregonensis*) au filet-trappe dans les lacs Griffen et Cultus, Colombie-Britannique, indiquent des pics d'activité et d'alimentation aux heures de demi-jour et de noirceur. Au lac Griffen, des sauvagesses du nord gavées de 2.0 g de menés roses (*Richardsonius balteatus*) et maintenues à 17-20 C ont un taux de digestion d'environ 14%/h. Au lac Cultus, des sauvagesses du nord capturées dans des trappes au moment du pic d'alimentation et maintenues à 10-12 C ont été tuées à intervalles de 6 h et ont digéré des smolts de saumon nerka (*Oncorhynchus nerka*) au taux de 14%/h. Les grands poissons absorbent un plus grand volume de nourriture que les petits. Les calculs de régression ne sont pas tout à fait adéquats pour décrire le cours de la digestion qui, semble-t-il, comporte une phase initiale de rétention des aliments, suivie d'une rapide évacuation de la plus grande partie du matériel ingéré. La plupart des poissons ont un transit intestinal de moins de 24 h, et la moitié entre 12 et 18 h. Au laboratoire, des sauvagesses du nord gavées de truites arc-en-ciel (*Salmo gairdneri*) ont un taux de digestion qui passe d'environ 5%/h à 4-6 C à 40-50%/h à 24 C. La digestion est plus lente lorsque le poids ingéré est plus grand, mais il n'y a pas de différence uniforme dans le taux de digestion de sauvagesses du nord de différentes tailles.

Les taux de digestion sont plus élevés que ceux qui ont été signalés chez divers poissons prédateurs d'Europe.

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THE squawfish (*Ptychocheilus oregonensis*) is well known as a predator of juvenile salmon and other fishes in Pacific drainages (Clemens and Munro

1934; Ricker 1941; J. W. Cartwright, University of British Columbia, Vancouver, B.C., unpublished data; Thompson 1959). The extent of predation is not clear. For example, Ricker (1941) estimated consumption on the basis of quantities of food

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found in the stomachs of gillnet-caught fish, though it was not known how long the fish had been in the nets and how far digestion had proceeded. Thompson (1959) used gillnet samples to assess the role of squawfish as predators on salmon in the Lower Columbia River but, again, rates of digestion were not known. The studies reported in this paper were aimed at providing information on rates of digestion of squawfish of various sizes when fed on various quantities of fish, and held at various temperatures. Initially, experiments were conducted in field conditions at Griffen Lake, near Revelstoke, B.C. Subsequently, laboratory and field studies were done at Cultus Lake, near Chilliwack, B.C., with the assistance of the International Pacific Salmon Fisheries Commission. Additionally, observations were made at both lakes on the daily activity patterns of squawfish as indicated by net catches.

Feeding Activity

Griffen Lake — Squawfish were taken in a live trap set June 29, lifted September 4, 1968, and usually

cleared of fish daily or more frequently. The net was set from shore with a lead of 40 m, and fished to a depth of 10 m. The total catch was approximately 5000 fish, from which a sample of 874 was examined for stomach contents. Food was found in the stomachs of 497 fish, and fish remains in 328. Those containing fish were above 12.5-cm fork length, and substantial quantities of fish were present only in squawfish over 17.5 cm. The proportion of squawfish containing fish in their stomachs (Y) was related to fork length (X) by the regression: $Y = 0.0211 + 0.0171X$. Insect remains predominated in squawfish under 17.5 cm.

The natural logarithm of average weight¹ of fish in the stomachs (Y) increased linearly with fork length (centimeters) of the squawfish (X) according to the relation $Y = -0.6940 + 0.0627X$. Additionally, a total of 85 spawning squawfish was examined and, almost without exception, their stomachs contained only trace amounts of food.

There was no significant difference in the average weight of fish in the stomach over the 24-h daily period, but the variability in weight was far greater at 2100 h

¹Measured volumetrically, specific gravity assumed to be 1.

TABLE 1. Results of analysis of stomach contents of northern squawfish (*Ptychocheilus oregonensis*) from Griffen Lake (July–September, 1967).

Food items	No. fish containing food items at different times					Totals
	0900	1300	2100	0100	0500	
Fish	269	34	4	4	17	328
Insect	39	11	0	2	28	80
Other	59	11	2	2	15	89
Empty	239	78	13	8	39	377
Total	606	134	19	16	99	874
Avg wt fish (g) and SD	3.47 ± .41	2.32 ± 1.45	3.06 ± 6.39	7.25 ± 9.01	1.29 ± 1.13	
% occurrence fish	44.39	25.37	21.05	25.00	17.17	
% occurrence insects	6.44	8.20	0.00	12.50	28.28	

TABLE 2. Summary of the trap-catch data collected at Griffen Lake, August 20–25, 1967.

	Time									
	0100 <i>a</i>	0500 <i>b</i>	0900 <i>c</i>	1300 <i>d</i>	1700 <i>e</i>	2100 <i>f</i>	0100 <i>a</i>	0500 <i>a</i>		
Aug. 20	:	33(B ₁)	:	3(D ₁)	:	132(A ₁)	:			
Aug. 21	:	84(B ₂)	:	5(D ₂)	:	74(A ₂)	:			
Aug. 22	:	58(B ₃)	:		26(F ₁)	:	67(C ₁)	:		
Aug. 23	:	:		27(G ₁)	:	:	18(C ₂)	:		
Aug. 24	:	:	21(E ₁)		:	40(A ₃)	:			
Aug. 25	:	37(B ₄)	:							

and 0100 h than at other times (Table 1), circumstantially suggesting that feeding occurs predominantly from dusk to dawn and the trap was catching fish during the feeding period.

If it can be assumed that greater activity was associated with feeding, and that trap catches reflected activity, it was evident from catches that feeding was occurring in the twilight and dark hours because overnight catches were almost invariably larger than those in daytime. To better delineate periods of activity, the trap was lifted at prescribed intervals on three occasions. July 26–28 the trap was lifted at 0500, 1300, and 2100 h; August 2–5 the lifts were at 0900, 1300, and 2100 h. In both cases the peak catch was at the morning lift.

Lifting of the trap at too frequent intervals appeared to disrupt the catching process, so to further pinpoint the hours of greatest catches a special pattern of setting was devised that enabled deduction of the average numbers of fish taken in different time periods. The intervals for recording the number in the trap, designated as capital letters, and 4-h time intervals in the day (small letters), may be represented as follows:

Time	Catch in interval in day	Setting interval
0100	$\overline{a}$	$\overline{B}$
0500	$\overline{b}$	$\overline{B}$
0900	$\overline{c}$	$\overline{B}$
1300	$\overline{d}$	$\overline{D}$
1700	$\overline{e}$	$\overline{D}$
2100	$\overline{f}$	$\overline{A}$
0100	$\overline{f}$	$\overline{A}$
0500	$\overline{a}$	$\overline{B}$

Replicates of setting intervals are denoted by subscripts (e.g. A₁, A₂, A₃). The sequence of the setting intervals over a 6-day period was:

Aug. 20—0100—B₁—0900—D₁—1700—A₁
 Aug. 21—A₁—0100—B₂—0900—D₂—1700—A₂
 Aug. 22—A₂—0100—B₃—0900—F₁—2100—C₁
 Aug. 23—C₁—0500—G₁—2100—C₂
 Aug. 24—C₂—0500—E₁—1700—A₃
 Aug. 25—A₃—0100—B₄—0900

The catches in the various intervals are given in Table 2. This arrangement partitions the darker hours of the day into overlapping 4-h time periods. The solution for the determination of average catches in each period proceeds on the premise that all first deductions are used to provide average values before proceeding to second and further deductions; i.e. from Table 2 three estimates of catch in the interval $\overline{e}$ are available:

F₁–D₁, F₁–D₂, G₁–E₁; and these are, respectively, 23, 21, and 6, with a mean $\overline{e} = 17$. Similarly, three estimates of $\overline{b}$ are available: E₁–D₁, E₁–D₂, G₁–F₁; and these are, respectively, 18, 16, and 1, with a mean $\overline{b} = 12$. From $\overline{b} = 12$ there are four estimates of catch in interval $\overline{a}$ available from subtracting $\overline{b}$ from each of the four B values, and these are, respectively, 23, 74, 48, and 27, for a mean $\overline{a}$ of 43. From three estimates of $\overline{f}$ are available from subtracting $\overline{e}$ from each of the A values, and the estimates of $\overline{f}$ are, respectively, 115, 57, and 23, for a mean $\overline{f}$ of 65. Finally, from E– $\overline{b}$ and F– $\overline{e}$ we can deduce ($\overline{c} + \overline{d}$) as 9 and 9, and these values averaged with D₁ and D₂ (3 and 5) yield ($\overline{c} + \overline{d}$) = 7.

Thus, the average catches, and presumably the activity of fish at different times, can be summarized as in Table 3, with a peak at 2100–0100 h and most of the remainder of the catch at 0100–0500 h.

Cultus Lake—A trapnet was set throughout the period April 20–June 3, 1969, in Cultus Lake near the

TABLE 3. Deduced number of squawfish taken at various time periods at Griffen Lake.

Time period	Activity index
$\overline{a} = 0100-0500$	43
$\overline{b} = 0500-0900$	12
$\overline{c} + \overline{d} = 0900-1700$	7
$\overline{e} = 1700-2100$	17
$\overline{f} = 2100-0100$	65

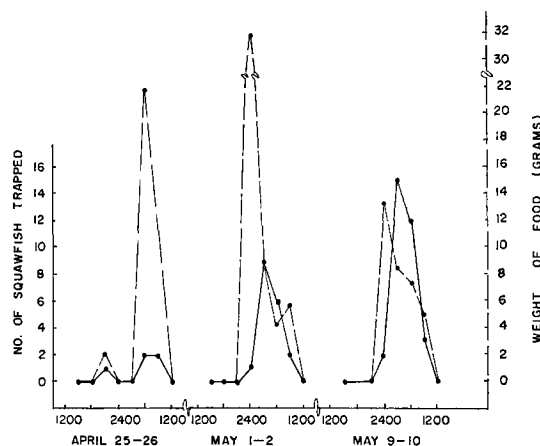


FIG. 1. Catch (dashed line) and weight of food in stomachs (solid line) of northern squawfish (*Ptychocheilus oregonensis*) from a trap net lifted at 3-h intervals from 1500 h onwards on April 25–26, May 1–2, and May 9–10, 1969, Cultus Lake.

outlet stream. On three occasions, April 25-26, May 1-2, and May 9-10, during the peak period of smolt migration, the trapnet was lifted at 3-h intervals. The greatest numbers of squawfish were caught between 2400 and 0600 h, and the mean volume of food in the stomachs was also highest at these times (Fig. 1). The squawfish were consuming sockeye salmon smolts almost exclusively, only 15% had empty stomachs, and the average number of smolts consumed was between two and three on all three occasions (mean number 2.50, mean weight 10.51 g, sample size 55). As in the case of the Griffen Lake experiments, peak feeding activity seemed to occur during the hours of darkness, in Cultus Lake the time of feeding no doubt reflecting in some measure the daily pattern of smolt migration.

Rate of Digestion

Griffen Lake — As a first digestion experiment, 164 squawfish ranging in size from 19.4- to 38.3-cm fork length (mean 24 cm) were force-fed 2.0 ± 0.1 g of reidside shiner (*Richardsonius balteatus*). This quantity

of food was somewhat below the average found in stomachs of fish from the trap, but was a convenient amount because of the size of the shiners and the rigors of force-feeding. The squawfish were held in herring mesh enclosures $2 \times 2 \times 3$ m deep and starved for 72 h before feeding. Immediately prior to feeding, the fish were anesthetized with tricane methanesulfonate (Sandoz M.S. 222) at a concentration of 1-15,000. The residual stomach contents were pumped out at 6, 12, 18, or 24 h after feeding, the fish being anesthetized again for this operation. The stomach pump was modelled after that of Seaburg (1957), and the pumped residues were filtered through screen of 1-mm mesh before weighing to 0.1-g accuracy. During the course of the experiments the ambient water temperature varied between 17 and 20 C.

The rate of digestion was best described by a regression of log weight of food remaining² on time in

²In calculating this regression, 0.01 was added to each weight of food remaining, to avoid the generation of negative infinite log values.

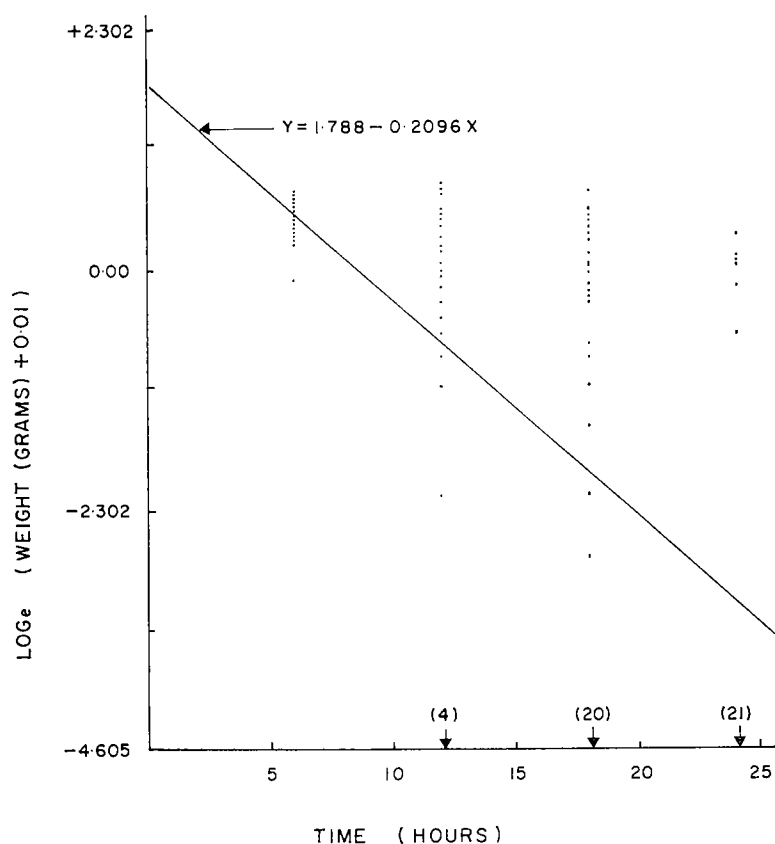


FIG. 2. Regression of natural logarithm of weight of food in stomach against time, for 164 squawfish fed 2.0 g of reidside shiner (*Richardsonius balteatus*). Numbers of fish with empty stomachs given in brackets. Griffen Lake, June 1968.

hours (Fig. 2), but the wide scatter of the data suggested that the circumstances of the experiment were not adequately controlled. Among the contributing factors to the scatter were the variable length of the fish, varying temperature, the effects of the anesthetic, force-feeding and handling, and the use of wet weights. For simple regression, the instantaneous rate of digestion was 0.2096, or roughly 20%/h, but the intercept of the regression (1.788) was a poor estimate of the log initial weight fed (0.698). When the regression was constrained to go through the known log weight of food that was initially fed (0.698), the estimate of the rate of digestion was 0.148, corresponding to roughly 14%/h.

Cultus Lake — (Field experiments) — Approximately 650 squawfish were removed from the Cultus Lake live trap between 1200 May 1 and 1000 May 2. The fish were divided into five groups. One of these groups was killed and preserved immediately, and the other groups were sacrificed after 6, 12, 18, and 24 h, respectively. The temperature at the capture site and in the holding pens ranged from 10 to 12 C.

The stomach contents were almost exclusively sockeye salmon smolts. The natural logarithm of the average volume of stomach contents (Y) at the time of capture was significantly linearly related to the

fork length of the fish (X) in a sample of 100 by the relation: $Y = -9.4130 + 0.4350X$. The regression of log volume of food in the stomach against time since capture was undertaken separately for each of 10 size-groups at 1-cm intervals from 24.5 to 34.5 cm. The slopes of the regressions were not significantly different,

TABLE 4. Regression statistics for the relation between $\log_e$ weight of food in the stomach (Y) and time in hours since capture (X) of squawfish of 10 ranges of fork length, Cultus Lake, May 1-2, 1969.

Fork length (cm)	Intercept	Slope	Sample size
24.6-25.5	0.9519	-0.1973	12
25.6-26.5	0.7191	-0.1384	28
26.6-27.5	-0.0762	-0.1504	42
27.6-28.5	-0.5731	-0.1245	82
28.6-29.5	-0.5381	-0.0847	104
29.6-30.5	0.6440	-0.1467	117
30.6-31.5	0.8899	-0.1439	97
31.6-32.5	1.6643	-0.1345	81
32.6-33.5	1.5736	-0.1464	50
33.6-34.5	1.9954	-0.1575	45

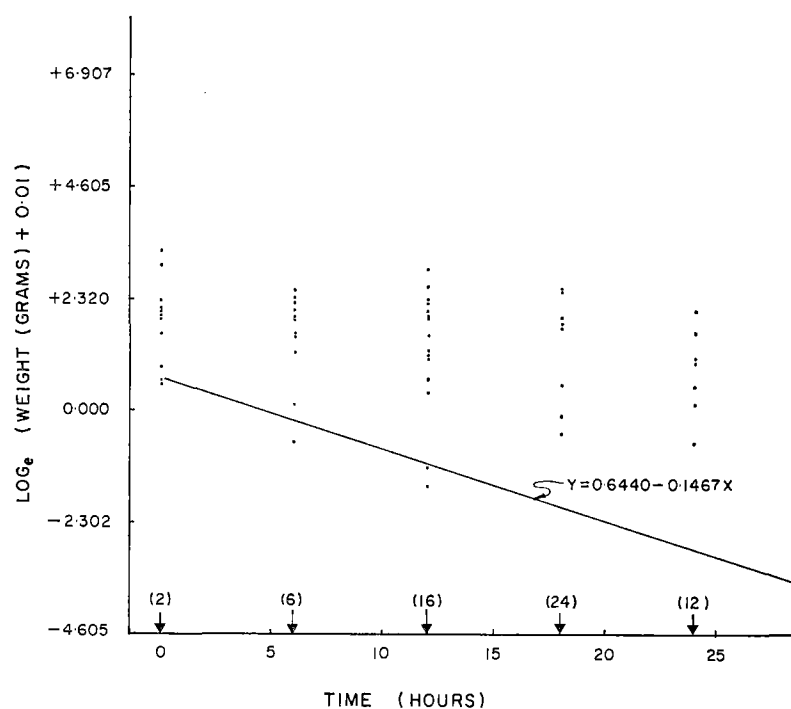


FIG. 3. Regression against time of the natural logarithm of weight of food in stomachs of 117 squawfish (29.6-30.5 cm) sacrificed at capture and at various times after capture. Numbers of fish with empty stomachs given in brackets. Cultus Lake, May 1-2, 1969.

and almost uniformly reflected a rate of digestion of about 14%/h for all size-groups. The intercepts were significantly different, reflecting the larger volumes of food in the stomachs of the larger fish (Table 4).

Inspection of the data suggests that the slope of the regression of log weight of food in the stomach against time since capture may not be entirely adequate as a measure of rate of digestion for individual squawfish. For all sizes of fish it was notable that at all times

relatively few fish contained small quantities of food in their stomachs. Figure 3 is typical, and illustrates that, at later times, the proportion of fish with empty stomachs increased from 16% at time of capture, to 63% after 24 h (Table 5). The simplest explanation for these results is that the food is retained virtually intact in the stomach during early stages of digestion and is then rapidly liquefied and passed to the intestine. Nevertheless, the majority of squawfish have cleared

TABLE 5. Frequency of occurrence at time of capture and after various lengths of time, of empty stomachs and stomachs containing food in squawfish of various fork lengths. Data collected at Cultus Lake, B.C., May 1-2, 1969.

Fork length (cm)	Time since capture									
	0 h		6 h		12 h		18 h		24 h	
	Empty	With food	Empty	With food	Empty	With food	Empty	With food	Empty	With food
24.6-25.5	0	1	0	2	1	1	2	1	3	1
25.6-26.5	0	1	0	3	3	4	2	2	8	5
26.6-27.5	0	6	4	2	7	7	6	2	6	2
27.6-28.5	7	9	2	5	11	11	10	6	18	3
28.6-29.5	3	6	7	6	17	20	11	9	15	10
29.6-30.5	2	16	6	13	16	21	14	8	12	9
30.6-31.5	3	15	4	11	8	17	11	9	12	7
31.6-32.5	0	16	2	13	4	14	6	10	9	7
32.6-33.5	1	9	1	11	4	8	1	4	6	5
33.6-34.5	0	5	2	6	2	10	5	5	5	5
Totals	16	84	28	72	73	113	68	56	94	54

TABLE 6. Cultus Lake laboratory experiments on rate of digestion in squawfish. Each replicate included three fish, one at each of three volumes of food.

Temp range (C)	Size categories ^a	No. of replicates	Dissection time (hours)							
			4	6	8	10	12	14	18	24
4-6	1	2		x			x		x	x
	2	2		x			x		x	x
	3	2		x			x		x	x
	4	2		x			x		x	x
	5	2		x			x		x	x
	6	2		x			x		x	x
8-10	2	2		x			x		x	x
	3	4		x			x		x	x
	4	4		x			x		x	x
15-16	2	2		x			x		x	x
	3	4		x			x		x	x
	4	4		x			x		x	x
19-21	3	4	x	x	x	x	x	x		
24-25	3	4-5	x	x	x	x	x			

^aCategories: 1 = Fork length (FL) 15.0-19.9 cm, 2 = (FL) 20.0-24.9, 3 = (FL) 25.0-29.9, 4 = (FL) 30.0-34.9, 5 = (FL) 35.0-39.9, 6 = (FL) >40.0.

TABLE 7. Regression statistics for relation between weight of food in the stomach and time since feeding for squawfish of various fork lengths held at various temperatures, Cultus Lake, June 1969. (Asterisk denotes significance at $\alpha = .05$.)

Fork length (cm)	Wt fed	Log (W +.01)	Temp 6 C			Temp 10 C			Temp 15 C			Temp 20 C			Temp 24 C		
			Sample size	Intercept	Slope	Sample size	Intercept	Slope	Sample size	Intercept	Slope	Sample size	Intercept	Slope	Sample size	Intercept	Slope
15.0-19.9																	
Pooled			21	0.7692	-0.03683												
Low	0.87	-0.128	8	0.5771	-0.02787*												
Medium	1.17	0.165	6	1.889	-0.1351												
High	3.46	1.244	7	1.271	-0.008517												
20.0-24.9																	
Pooled			21	1.060	-0.03728	22	1.369	-0.1236*	21	1.511	-0.02413*						
Low	1.22	0.207	8	0.2780	-0.03897	8	0.1971	-0.1212*	6	-1.162	-0.1501*						
Medium	2.39	0.875	8	0.7576	0.001719	8	1.874	-0.1511*	8	1.96	-0.2760*						
High	4.79	1.569	5	1.448	0.004693	6	2.077	-0.07777*	7	2.442	-0.2216*						
25.0-29.9																	
Pooled			23	1.330	-0.04367	45	1.052	-0.06710*	45	1.674	-0.2412*	68	1.993	-0.4143*	67	1.920	-0.4869*
Low	1.60	0.476	8	1.100	-0.08537*	16	0.08154	-0.08945*	16	-0.362	-0.1784*	24	0.2975	-0.3705*	22	0.4854	-0.4542*
Medium	3.19	1.163	8	1.180	-0.02745	15	1.726	-0.1103*	13	1.756	-0.2477*	24	2.574	-0.4504*	22	2.621	-0.5441*
High	6.39	1.860	7	1.913	-0.02492*	14	1.941	-0.03335*	16	3.647	-0.2994*	20	3.519	-0.4440*	23	3.025	-0.5116*
30.0-34.9																	
Pooled			23	1.551	-0.03576	45	1.246	-0.05744*	46	2.151	-0.02652*						
Low	2.00	0.698	8	0.8516	-0.04694	16	0.7458	-0.09576*	16	0.2923	-0.213*						
Medium	4.00	1.389	7	1.669	-0.04303	16	1.457	-0.05723*	15	2.586	-0.2775*						
High	8.00	2.081	8	2.169	-0.01917*	13	2.352	-0.06020*	15	3.539	-0.2990*						
35.0-39.9																	
Pooled			24	2.003	-0.06558*												
Low	2.42	0.888	8	1.801	-0.1098*												
Medium	4.84	1.579	8	1.715	-0.4174												
High	9.68	2.271	8	2.493	-0.04514												
>40.0																	
Pooled			20	3.906	-0.1403												
Low	2.86	1.054	7	1.815	-0.1201												
Medium	5.73	1.747	8	2.533	-0.09301*												
High	11.44	2.438	5	6.268	-0.08294												
Sizes pooled																	
Pooled				1.751	-0.05896*		1.169	-0.07247*		1.838	-0.2511*		1.993	-0.4143*		1.920	-0.4869*
Low				0.9285	-0.06574*		0.3704	-0.09834*		-0.1876	-0.1902*		0.2975	-0.3705*		0.4854	-0.4542*
Medium				1.554	-0.04937*		1.635	-0.09598*		2.119	-0.2653*		2.574	-0.4504*		2.621	-0.5441*
High				2.693	-0.04494		2.052	-0.04718*		3.385	-0.2859*		3.519	-0.4440*		3.025	-0.5116*

TABLE 8. Statistics for regression of weight of food in stomach and time after feeding when regression constrained through the known weight of food fed, Cultus Lake, June 1969. (Asterisk denotes significance at $\alpha = .05$.)

Fork length (cm)	Log ($W_t + .01$)	6 C		10 C		15 C		20 C		24 C	
		Sample size	Slope	Sample size	Slope	Sample size	Slope	Sample size	Slope	Sample size	Slope
15.0–19.9											
Pooled		21									
Low	-0.128	8	-0.01756*								
Medium	0.165	6	-0.05748								
High	1.244	7	-0.007057*								
20.0–24.9											
Pooled		21		22		21					
Low	0.207	8	-0.03502*	8	-0.1218*	6	-0.2298*				
Medium	0.875	8	-0.005057	8	-0.09591*	8	-0.2157*				
High	1.569	5	-0.004802	6	-0.04818*	7	-0.1687*				
25.0–29.9											
Pooled		23		45		45		68		67	
Low	0.476	8	-0.05067*	16	-0.1113*	16	-0.2249*	24	-0.3878*	22	-0.4532*
Medium	1.163	8	-0.02649*	15	-0.07778*	13	-0.2114*	24	-0.3133*	22	-0.3855*
High	1.860	7	-0.02185*	14	-0.02879*	16	-0.1999*	20	-0.2859*	23	-0.3838*
30.0–34.9											
Pooled		23		45		46					
Low	0.698	8	-0.03841*	16	-0.09311*	16	-0.2355*				
Medium	1.389	7	-0.02795*	16	-0.05343*	15	-0.2121*				
High	2.081	8	-0.01425*	13	-0.04588*	15	-0.2149*				
35.0–39.9											
Pooled		24									
Low	0.888	8	-0.05912*								
Medium	1.579	8	-0.03415*								
High	2.271	8	-0.03280*								
> 40.0											
Pooled		20									
Low	1.054	7	-0.06125*								
Medium	1.747	8	-0.04936*								
High	2.438	5	+0.1435								

their stomach contents within 24 h, the median point lying between 12 and 18 h.

Laboratory experiments—To determine the effect of temperature, volume of food, and size of fish on the rate of digestion, squawfish of various sizes were force-fed known volumes of food and sacrificed after various intervals of time. It was not possible to do all combinations of treatments in a factorial design; Table 6 summarizes what was undertaken. Trapped squawfish were held in a large rearing channel (11–13 C) and, prior to the experimental period, were fed ad lib. on redeye shiners (*R. balteatus*). The evening prior to a feeding experiment, squawfish were transferred to plywood controlled temperature tanks $0.6 \times 2.6 \times 0.4$ m deep. After 12 h, they were force-fed one redeye shiner, and they were then starved for 72 h, at which time they were force-fed a prescribed weight of juvenile (3–6 cm) hatchery rainbow trout (*Salmo gairdneri*). At a prescribed time after the feeding the squawfish were killed, the stomach contents removed, washed over screening of 1-mm mesh, damp dried, and quick frozen for subsequent weighing.

The weights of food fed were arbitrarily set at low (2 g) medium (4 g), and high (8 g) for fish of 32.5-cm fork length, a range that spanned that observed at Griffen Lake in the stomachs of trapped fish (for 32.5-cm fish the mean weight of stomach contents was 3.84 g). For fish of other sizes the weight of food was chosen to be proportional to the 0.44 power of weight of the fish. Although this rule is contrary to the generally accepted rule that ration is proportional to the two-thirds power of weight, it provided rations that approximated what were observed in the field at Griffen Lake (e.g. for a fish of 12.5-cm fork length, 1.06 g; 22.5-cm, 3.84 g; 40-cm, 6.14 g). The weights of food on the stomachs of trap-caught fish at Cultus Lake were also in the same range as those fed in the laboratory. For example, size 3 fish (25.0–29.9 cm) taken in the field contained an average stomach content of 3.76 g of food, slightly more than the medium ration in the laboratory experiment.

The results are summarized in Table 7, giving slopes and intercepts for standard regression, and Table 8, slopes for regressions constrained through the known weight of food fed. Contrasting the results in the two tables, standard regression techniques usually overestimated the amount of food fed (intercept greater than known weight fed) and a higher estimate of the rate of digestion than constrained regression. Apparently, this results from an initial period of hydrolysis of the food, which increases its weight, followed by a period in which the food is retained in the stomach during early stages of digestion, and is then followed by a short period of relatively fast evacuation of most of the stomach contents. Because the period of observation ended shortly after small residues remained, the initially quantity of food fed was overestimated. To a lesser degree the average rate of digestion was also overestimated.

Despite the differences in the two analyses, they both show three major effects: (1) the rate of digestion

substantially increases with temperature, from about 5%/h at 4–6 C, to 40–50%/h at 24 C; (2) larger weights of food are digested at slower rates; and (3) there are no consistent differences in the rate of digestion by different sizes of fish. Figure 4 illustrates the substantial increase in rate of digestion with increased temperature for size 3 fish, and the consistently higher rates of digestion of smaller quantities of food.

Discussion

The foregoing results clearly indicate that squawfish feed most intensively in the hours of dusk and darkness, and that an average meal is usually evacuated from the stomach within 24 h. The ambient temperature where squawfish are most commonly taken in nets approximates 10–15 C at which digestion rates for various volumes of food

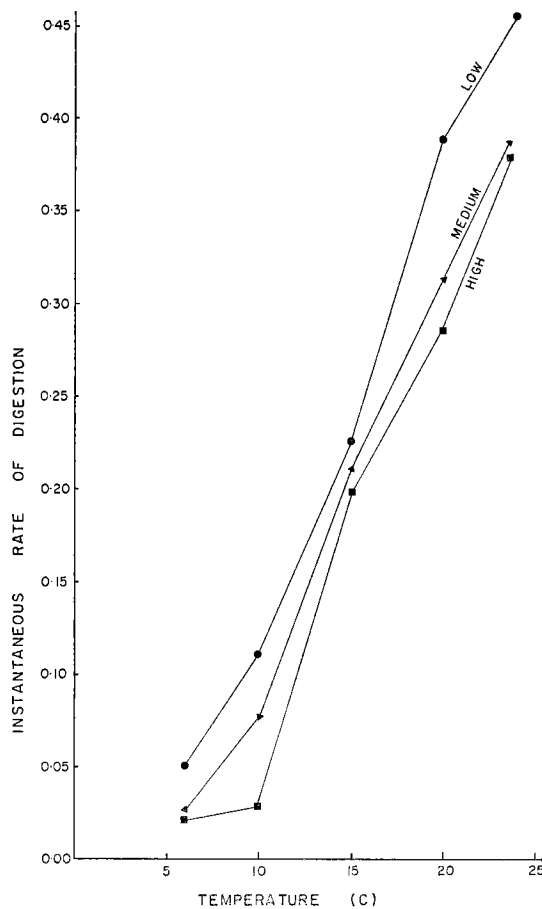


FIG. 4. Instantaneous hourly rate of digestion in stomachs of squawfish of fork length 25.0–29.9 cm at various temperatures for low (1.60 g), medium (3.19 g), and high (6.39 g) levels of food.

are in the range of 5–20%/h (Table 8). This rate is not dissimilar to that for bluegill sunfish (*Lepomis macrochirus*) at 18–21 C, as calculated from the results of Windell (1967, figure 1), for which several kinds of foods were digested at approximately 13 %/h. The stomachs of bluegill sunfish were usually empty from 17–19 h after a natural feeding.

All the rates of digestion here reported for squawfish are much higher than those given by Molnár et al. (1967) for various species of predatory fishes of comparable size and at comparable temperature, although in Molnár's work the size of fish used as food was larger—sufficient to fill the stomach. For example, at 10 C, gastric digestion in largemouth bass (*Micropterus salmoides*) was not completed for 50 h. The digestion rates in excess of 40%/h for squawfish held at 24 C in our laboratory experiments are extraordinary, but were observed consistently for various weights of food and were part of a trend to higher rates at higher temperatures. They are far in excess of rates reported by Molnár et al. (1967); for example, at 25 C pikeperch (*Lucioperca lucioperca*), Eurasian perch (*Perca fluviatilis*), and largemouth bass (*M. salmoides*) did not achieve 100% gastric digestion until 28, 21, and 19 h, respectively. The change in rate of digestion with temperature is comparable to that reported by Molnár et al. for pike, perch, and the European catfish (*Silurus glanis*).

The findings of this study have implications for field assessment of total consumption of fish by squawfish. For example, if squawfish are left in gillnets for extended periods after they are captured, the quantity of food in the stomach when the nets are lifted may be a poor indicator of what they take as an average meal. Without knowledge of when the fish were caught, and the length of time that squawfish remain alive (and digesting) in nets after capture, it is difficult to be precise about what "correction factor" might be applied, but it is obvious that estimates of consumption based on gillnet-caught fish are minimal. Thus, the estimates of Ricker (1941), in some years based on fish that had been taken in nets set for 48–72 h, are undoubtedly low. Even when nets were set for 24 h, it would be anticipated that substantial digestion would have occurred, and though Ricker was aware of this source of error, it would appear to have been greater than he appreciated. The lower temperatures of the early spring period when squawfish predation on sockeye fry and smolts is intense, are in some

measure associated with slower digestion rates, but even at 4–6 C, the rate of digestion approximates 3%/h and enables two-thirds evacuation of the stomach within 36 h.

Average rates of digestion may be misleading. In laboratory studies, standard regression of log weight of food fed against time overestimated the amount of food fed, reflecting an initial period of hydrolysis followed by a fast rate of evacuation of most of the stomach contents. The same phenomenon would account for the field observations of digestion rate, for which the average time to evacuation of the stomach was more meaningful than the average digestion rate. Under field conditions, with ambient temperatures of 10–12 C, and with feeding natural, two-thirds of the squawfish had empty stomachs after 24 h.

The merits of control of predators such as squawfish, for increasing production of sockeye salmon, have been a matter of anecdotal remark for many years. The estimates of Ricker (1941) and Foerster and Ricker (1941) indicated that, because of the quantity of prey taken, squawfish removal programs could be profitable. This study suggests that the total consumption of sockeye by squawfish could be even larger than these authors estimated, and the benefits of control programs might thus be accordingly greater.

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